

Air University



Network Security (Lab)

Year 2026

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Academic Year: 2024 – 2028

Department of Cyber Security

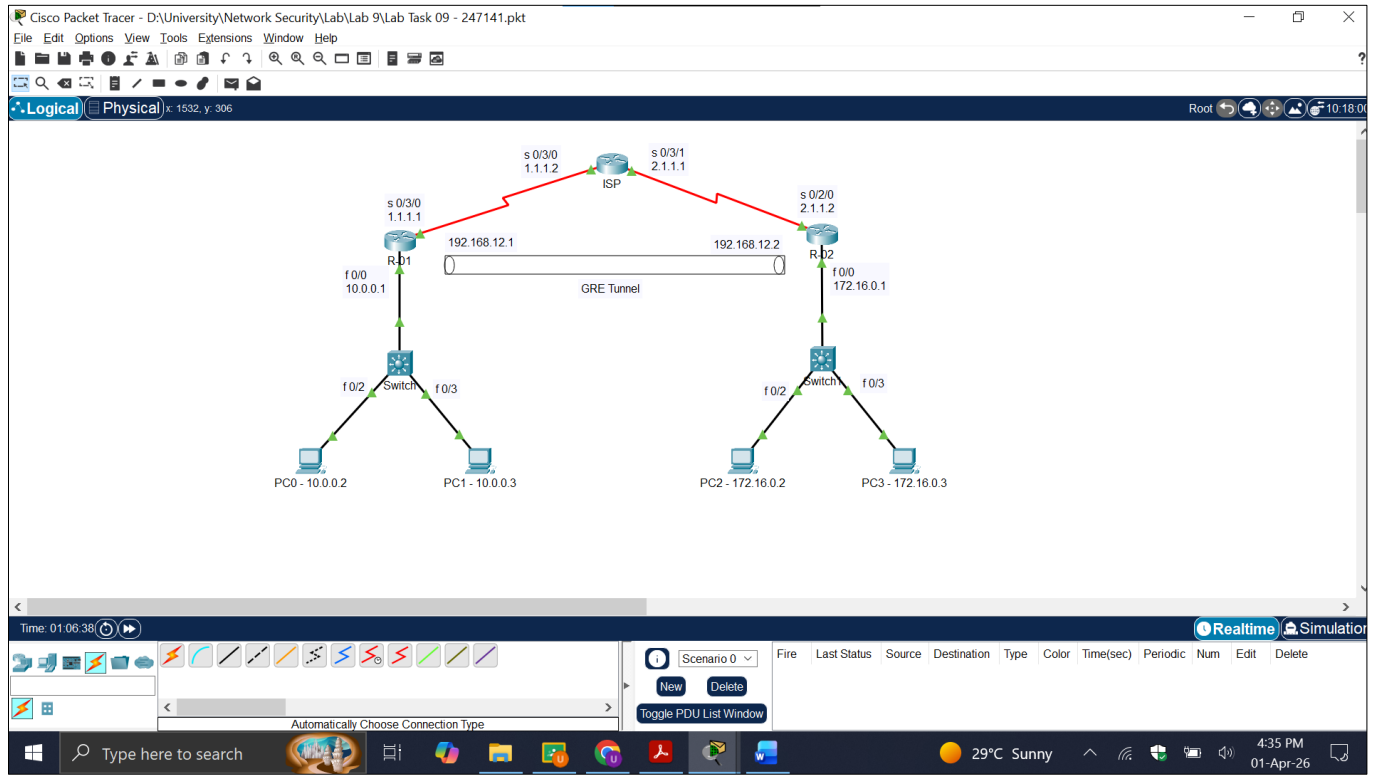
USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: April 01, 2026

Lab # 09

Topology:



Assigning IP Address:

Router-1

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-1
Router-1(config)#int s0/30/
^
% Invalid input detected at '^' marker.
Router-1(config)#int s0/3/0
Router-1(config-if)#ip address 1.1.1.1 255.0.0.0
Router-1(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router-1(config-if)#exit
Router-1(config)#int f 0/0
Router-1(config-if)#ip address 10.0.0.1 255.0.0.0
^
% Invalid input detected at '^' marker.
Router-1(config-if)#ip address 10.0.0.1 255.0.0.0
Router-1(config-if)#no shutdown

Router-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router-1(config-if)#exit
Router-1(config)#exit
Router-1#
%SYS-5-CONFIG_I: Configured from console by console

Router-1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.0.0.1        YES manual  up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Serial0/3/0     1.1.1.1         YES manual  down        down
Vlan1          unassigned      YES unset  administratively down down
Router-1#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
```

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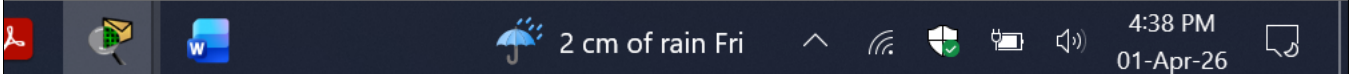
Top

```
Router-1>en
Router-1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-1(config)#int s 0/3/0
Router-1(config-if)#encapsulation ppp
Router-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down

Router-1(config-if)#bandwidth 1000
Router-1(config-if)#clock rate 64000
Router-1(config-if)#no shutdown
Router-1(config-if)#exit
Router-1(config)#exit
Router-1#
%SYS-5-CONFIG_I: Configured from console by console
```

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Router ISP

```
Router>en
Router#congiti
Translating "congiti"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#int s0/3/0
ISP(config-if)#ip address 1.1.1.2 255.0.0.0
ISP(config-if)#no shutdown

ISP(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

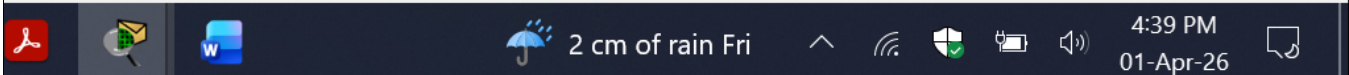
ISP(config-if)#exit
ISP(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

ISP(config)#int s 0/3/1
ISP(config-if)#
ISP(config-if)#ip address 2.1.1.1 255.0.0.0
ISP(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/1, changed state to down
ISP(config-if)#exit
ISP(config)#
```

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```
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to down

ISP(config)#int 0/3/0
      ^
% Invalid input detected at '^' marker.

ISP(config)#int s 0/3/0
ISP(config-if)#bandwidth 1000
ISP(config-if)#encapsulation ppp
ISP(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

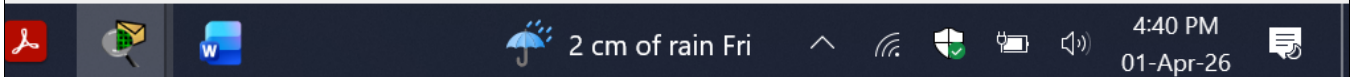
ISP(config-if)#no shutdown
ISP(config-if)#exit
ISP(config)#int s 0/3/1
ISP(config-if)#bandwidth 1000
ISP(config-if)#encapsulation ppp
ISP(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

ISP(config-if)#no shutdown
ISP(config-if)#exit
ISP(config)#
```

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Router-2

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Router-2
Router-2(config)#int s 0/2/0
Router-2(config-if)#ip address 2.1.1.2 255.0.0.0
Router-2(config-if)#no shutdown

Router-2(config-if)#
%LINK-5-CHANGED: Interface Serial0/2/0, changed state to up

Router-2(config-if)#exit
Router-2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up

Router-2(config)#int f 0/0
Router-2(config-if)#ip address 172.16.0.1 255.255.0.0
Router-2(config-if)#no shutdown

Router-2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router-2(config-if)#exit
Router-2(config)#int s 0/2/0
Router-2(config-if)#encapsulation ppp
Router-2(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to down

Router-2(config-if)#bandwidth 100
Router-2(config-if)#clockrate 64000
      ^
% Invalid input detected at '^' marker.

Router-2(config-if)#clock rate 64000
Router-2(config-if)#no shutdown
Router-2(config-if)#exit
Router-2(config)#exit
Router-2#
%SYS-5-CONFIG_I: Configured from console by console
```

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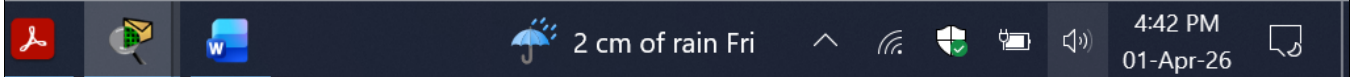


```
Router-2#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 172.16.0.1      YES manual up          up
FastEthernet0/1 unassigned      YES unset   administratively down down
Serial0/2/0     2.1.1.2        YES manual up          down
Vlan1          unassigned      YES unset   administratively down down
Router-2#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/2/0, changed state to up
```

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Configuration of GRE tunnel:

Router-1

```
Router-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#int tunnel 0

Router-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

Router-1(config-if)#tunnel source s 0/3/0
Router-1(config-if)#tunnel destination 2.1.1.2
Router-1(config-if)#ip address 192.168.12.1 255.255.255.0
Router-1(config-if)#exit
Router-1(config)#
Router-1(config)#ip route 2.0.0.0 255.0.0.0 1.1.1.2
Router-1(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

Router-1(config)#exit
Router-1#
%SYS-5-CONFIG_I: Configured from console by console

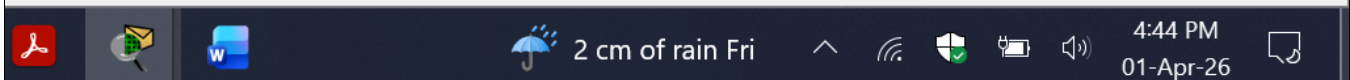
Router-1#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 10.0.0.1        YES manual up          up
FastEthernet0/1 unassigned      YES unset   administratively down down
Serial0/3/0     1.1.1.1         YES manual up          up
Tunnel0        192.168.12.1    YES manual up          up
Vlan1          unassigned      YES unset   administratively down down
Router-1#ping 192.168.12.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/8/30 ms
```

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Router-2

```
Router-2#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router-2(config)#int tunnel 0

Router-2(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

Router-2(config-if)#tunnel source s 0/2/0
Router-2(config-if)#tunnel destination 1.1.1.1
Router-2(config-if)#ip address 192.168.12.2 255.255.255.0
Router-2(config-if)#exit
Router-2(config)#
Router-2(config)#ip route 1.0.0.0 255.0.0.0 2.1.1.1
Router-2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

Router-2(config)#exit
Router-2#
%SYS-5-CONFIG_I: Configured from console by console

Router-2#ping 192.168.12.1
Translating "192.168.12.1"...domain server (255.255.255.255)
% Unrecognized host or address or protocol not running.

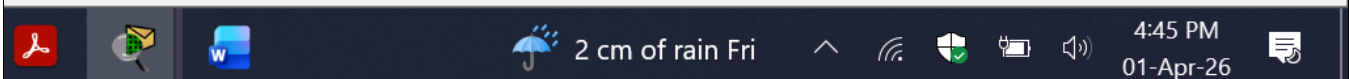
Router-2#ping 192.168.12.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.12.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/7/27 ms

Router-2#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 172.16.0.1      YES manual up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Serial0/2/0     2.1.1.2         YES manual up          up
Tunnel0        192.168.12.2    YES manual up          up
Vlan1          unassigned      YES unset  administratively down down
Router-2#
```

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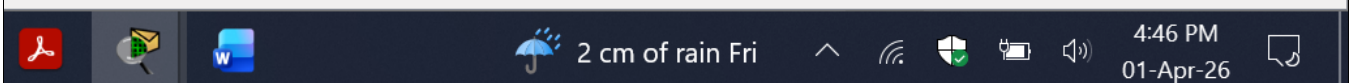
Static Routes:

Router-1

```
Router-1(config)#ip route 2.0.0.0 255.0.0.0 1.1.1.2
Router-1(config)#
```

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Router-2

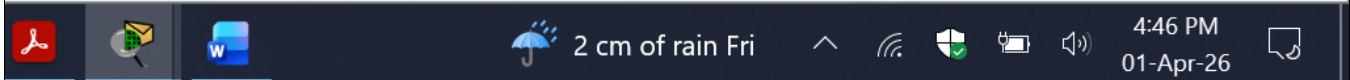
```
Router-2(config)#ip route 1.0.0.0 255.0.0.0 2.1.1.1
Router-2(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
```

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Paste to CLI

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EIGRP configuration:

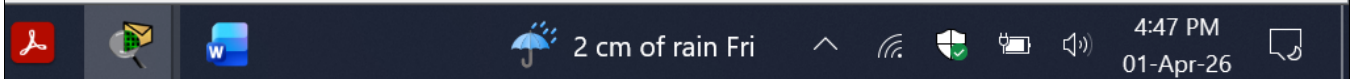
Router-1

```
Router-1#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-1(config)#router eigrp 100
Router-1(config-router)#network 10.0.0.0
Router-1(config-router)#network 192.168.12.0
Router-1(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.12.2 (Tunnel0) is up: new adjacency
```

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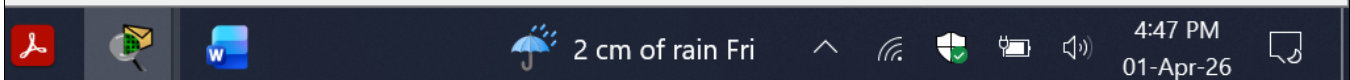
Router-2

```
Router-2#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router-2(config)#router eigrp 100
Router-2(config-router)#network 172.16.0.0
Router-2(config-router)#network 192.168.12.0
Router-2(config-router)#exit
Router-2(config)#
%DUAL-5-NBRCHANGE: IP-EIGRP 100: Neighbor 192.168.12.1 (Tunnel0) is up: new adjacency
```

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Packet Transfer:

RealtimeSimulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0 - 10.0.0.2	PC2 - 172.16.0.2	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC0 - 10.0.0.2	PC3 - 172.16.0.3	ICMP		0.000	N	1	(edit)	(delete)

< >

 2 cm of rain Fri      4:57 PM
01-Apr-26 

- End of Lab Task -